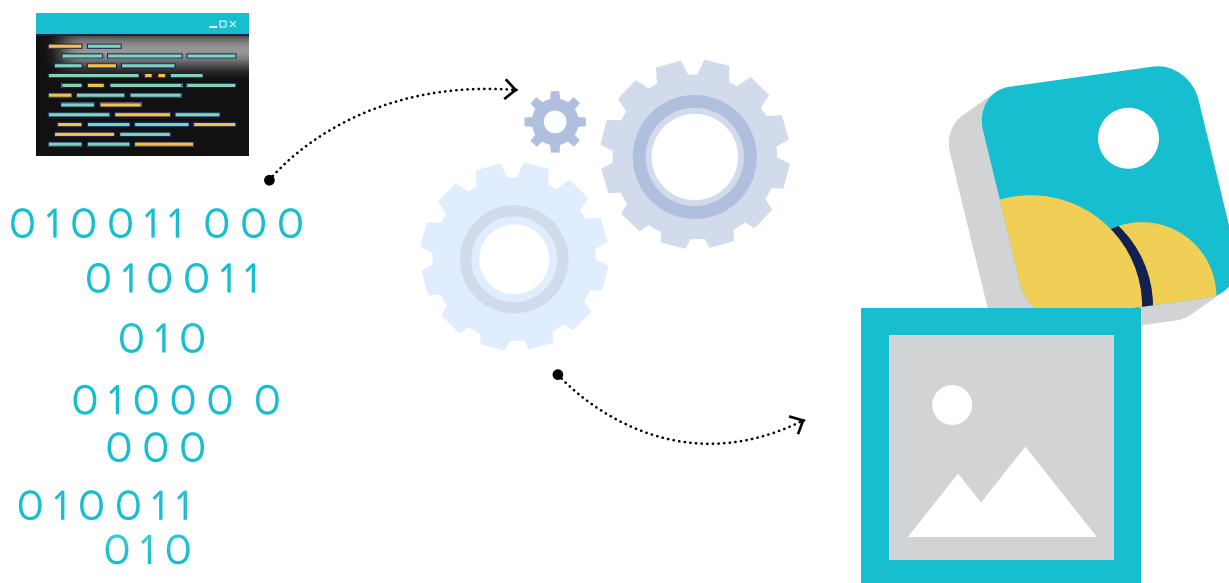


Word Embeddings Made Easy with Word2Vec

This short tutorial will teach you how to create your own Word2Vec model from a small corpus using the Gensim library.



Large language models (LLMs) like ChatGPT and Gemini are being used extensively today in the world of generative AI. We must therefore understand and learn the important concepts of natural language processing (NLP), which is the base for GenAI. One such concept is 'word embeddings'. While training a machine learning model, you need to give the inputs and outputs as numbers to the machine, because that is what it understands. Your machine does not understand text or image data directly. For any application of NLP, you must convert your text into meaningful numbers, which are called embeddings.

One such method to convert text data into numbers or numerical data is

Word2Vec. It literally means converting words to vectors, which are arrays/sets of numbers that represent a given word. A few other methods were used earlier to convert words to numbers (like one-hot encoding and Bag of Words). Word2Vec is better because the earlier methods could not capture the meaning of the order of text. Text is sequential data in which the order of words is important. Not capturing this information becomes a problem in many use cases.

Word2Vec models are trained on large corpuses to make them more useful. Once trained, these models can be used for a multitude of use cases like predicting the next word in a piece of text, detecting synonyms or predicting

how close a word is to another given word and, of course, to convert text into embeddings. This technique was developed by researchers at Google and was published in 2013. Word2Vec models are typically artificial neural network models, which create vectors that have dimensions in the hundreds. Each word in the corpus is assigned a unique vector. This is done using two methods -- CBOW or Continuous Bag of Words and Skipgram.

Let's now learn how to create your own Word2Vec model from a small corpus. We are going to use the Gensim library for this purpose. You can install this on your Jupyter Notebook using the following command as shown in Figure 1.